

Prefix-Outcome Validation for On-Policy Distillation of Reasoning Models

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Abstract

On-policy distillation (OPD) trains a student on its own sampled trajectories using dense token-level preferences from a stronger teacher. This provides fine-grained supervision on student-visited states, but it also assumes that every teacher logit difference is reliable along the whole sampled trajectory. In long-horizon reasoning, this assumption can fail: a local teacher preference may conflict with the final correctness of the response, and later prefixes may drift into regions where teacher guidance is less trustworthy. We propose *Prefix-Outcome Validation* (POV), a validation layer for OPD that calibrates dense teacher supervision using two complementary signals. Outcome validation uses final correctness to determine whether the teacher–student log-probability gap of each sampled token is aligned with the trajectory result, preserving aligned tokens while assigning only beta-capped rank weights to anti-aligned tokens. Prefix validation monitors teacher excess surprisal along the generated prefix and monotonically decays token supervision after sustained prefix drift. POV preserves OPD’s dense supervision while grounding it in both final outcome consistency and prefix-level teacher support, yielding a conservative alternative to outcome-only RL and unvalidated dense distillation. Experiments across multiple reasoning benchmarks demonstrate that POV provides more reliable credit assignment than OPD.

Introduction

Outcome-based reinforcement learning has become a standard post-training tool for improving the reasoning ability of language models. In mathematical reasoning, a verifier can often judge whether a completed response is correct, and policy optimization can then increase the probability of successful responses while decreasing the probability of failed ones (Schulman et al. 2017; Ouyang et al. 2022; Shao et al. 2024; DeepSeek-AI 2025). Group-relative methods such as GRPO make this setup especially simple: several responses are sampled for the same prompt, a scalar outcome reward is computed for each response, and the resulting relative advantage is broadcast to all response tokens. This design uses reliable final feedback, but it provides little information about which intermediate tokens should be reinforced.

On-policy distillation (OPD) offers a complementary source of supervision. Instead of training only from a scalar outcome, the student first samples its own response and a stronger teacher is then evaluated on the prefixes ac-

tually visited by the student. The teacher provides dense token-level guidance, often through the log-probability gap between the teacher and the student. This keeps training on-policy while exposing the student to fine-grained teacher preferences. For long reasoning traces, however, dense teacher supervision is not uniformly reliable. The teacher may locally prefer a token that belongs to an ultimately incorrect solution, or locally disfavor a token that appears in a correct trajectory. This mismatch creates a credit-assignment problem: the local teacher signal and the final outcome do not always agree.

A second issue arises from the prefixes on which the teacher is queried. OPD evaluates the teacher on student-generated prefixes rather than on teacher-generated ones. Once a student trajectory drifts into a low-support region under the teacher, later teacher logit differences still exist, but they may be less informative about how to reach a correct solution. Treating these later dense rewards as uniformly trustworthy can amplify supervision from invalid or off-path prefixes. Thus, OPD needs not only token-outcome consistency, but also a way to judge whether the current prefix remains a valid context for teacher supervision.

We propose *Prefix-Outcome Validation* (POV), a simple validation mechanism for on-policy distillation. POV uses final correctness to validate the direction of dense teacher preferences, rather than replacing them with a sparse reward. In a correct trajectory, a sampled token is outcome-aligned when the teacher assigns it higher log-probability than the student. In an incorrect trajectory, the alignment direction is reversed: a sampled token is outcome-aligned when the teacher assigns it lower log-probability than the student, thereby discouraging behavior observed in a failed response. This rule converts the final verifier result into a token-level reliability test for OPD supervision.

POV also validates prefixes. Since the teacher is queried on student-generated prefixes, the later parts of a response may drift into regions where the teacher’s dense preferences are less trustworthy. POV monitors teacher excess surprisal, defined as the negative teacher log-probability of the sampled token minus the teacher entropy at the same prefix. A sustained increase in excess surprisal indicates that the sampled prefix has become unexpectedly unlikely under the teacher. POV detects this change with a windowed CUSUM statistic and monotonically decays subsequent dense super-

vision after the detected horizon.

The resulting method, POV-OPD, keeps the useful part of dense OPD while reducing the influence of locally unreliable signals. Outcome validation assigns full weight to outcome-aligned sampled tokens and uses a beta-capped rank weight for anti-aligned tokens. Prefix validation then attenuates supervision after sustained teacher-surprisal drift. Thus, POV-OPD preserves token-level teacher guidance when it is consistent with both the final outcome and the teacher’s prefix-level support.

Our contributions are:

- We identify two reliability issues in on-policy distillation for reasoning: outcome-token mismatch and prefix drift under student-generated trajectories.
- We introduce outcome validation, which uses final correctness to partition sampled-token teacher preferences into outcome-aligned and anti-aligned signals.
- We introduce prefix validation, which detects sustained excess surprisal under the teacher and monotonically down-weights later dense supervision.
- We combine both mechanisms into POV-OPD, a conservative modification of dense OPD that preserves aligned supervision and assigns only beta-capped rank weight to anti-aligned tokens.

Related Work

Outcome-Based Reinforcement Learning for Reasoning

Reinforcement learning from outcome feedback is widely used to align language models and improve their reasoning behavior. PPO-style RLHF optimizes a policy against learned or programmatic rewards while controlling policy drift (Christiano et al. 2017; Schulman et al. 2017; Ouyang et al. 2022). Direct preference optimization and related preference-learning methods simplify parts of this pipeline when pairwise preference data are available (Rafailov et al. 2023). For mathematical reasoning, recent systems often use verifiable rewards, where a final answer can be checked automatically (Shao et al. 2024; DeepSeek-AI 2025). GRPO and related group-relative methods reduce the need for a learned value function by comparing multiple responses sampled for the same prompt and broadcasting a scalar relative advantage over response tokens (Shao et al. 2024). These methods use reliable final feedback, but they do not provide fine-grained token-level guidance. POV-OPD addresses this limitation by retaining dense teacher supervision while using final correctness as a validation signal.

Knowledge Distillation and On-Policy Distillation

Knowledge distillation transfers behavior from a stronger teacher model to a smaller or weaker student (Hinton, Vinyals, and Dean 2015). For sequence generation, distillation can imitate teacher outputs or teacher distributions at the token or sequence level. Offline distillation is simple, but it trains the student on states induced by the teacher or by a fixed dataset, which can differ from the states visited

by the student during reasoning. On-policy distillation reduces this mismatch by evaluating teacher preferences on student-generated trajectories. Recent work on OPD shows that dense token-level teacher rewards can be useful but sensitive to teacher-student compatibility and the states visited by the student (Li et al. 2026). POV-OPD follows the same on-policy view: the student samples the response, the teacher scores the visited prefixes, and the student updates from dense teacher-student log-probability gaps. The key difference is that POV-OPD does not assume every dense teacher preference is equally reliable.

Credit Assignment and Dense Supervision

Long-horizon reasoning requires assigning credit to intermediate decisions from sparse final outcomes. Chain-of-thought prompting, self-consistency, and self-training methods highlight the importance of intermediate reasoning paths for final-answer accuracy (Wei et al. 2022; Wang et al. 2022; Zelikman et al. 2022). Process supervision provides dense feedback for intermediate steps, but such supervision can be costly or noisy (Lightman et al. 2023). Dense teacher rewards offer an automatic alternative, yet they can conflict with the final correctness of the trajectory. POV-OPD treats this conflict as a validation problem: final correctness determines whether the local teacher-student preference points in an outcome-consistent direction. Our prefix validation is also related to classical change detection, where CUSUM statistics are used to detect sustained shifts in a monitored signal (Page 1954; Basseville and Nikiforov 1993). Here the monitored signal is teacher excess surprisal along a student-generated prefix, and the response is a monotone decay of later OPD weights rather than a hard rejection of the trajectory.

Limits of Outcome-Only and Prefix-Local Supervision

POV-OPD is motivated by two complementary limitations. A terminal outcome is reliable at the trajectory level but does not identify token-specific credit within one sampled response. Conversely, a teacher supplies dense token-level preferences, but global teacher superiority does not imply that every local preference remains reliable on student-generated prefixes. We formalize these two points below.

A Terminal Outcome Does Not Identify Token Credit

Let $Y = (y_1, \dots, y_T)$ be a response sampled from the student policy, and let $R(Y) \in \{0, 1\}$ be its verifier outcome. The ideal outcome credit of action y_t at prefix $h_t = (x, y_{<t})$ is its conditional advantage

$$A_t^R = Q^\pi(h_t, y_t) - V^\pi(h_t), \quad (1)$$

where $Q^\pi(h_t, y_t)$ is the expected final outcome after taking y_t and following π . Outcome-only methods instead construct a Monte Carlo estimate from the single terminal observation, typically

$$\hat{A}_t^{\text{out}} = R(Y) - b(x), \quad t = 1, \dots, T, \quad (2)$$

and broadcast the same scalar to every response token.

Proposition 1. For $T > 1$, the observed pair $(Y, R(Y))$ is insufficient to identify the token-level credits $\{A_t^R\}_{t=1}^T$ for an individual rollout.

Proof. Fix an observed rollout Y and its terminal reward $R(Y)$. Consider any position t and an alternative action y'_t . Two environments can assign exactly the same transitions and terminal reward along the observed rollout Y , while assigning different continuation rewards after the unobserved alternative y'_t . These environments therefore produce the same observation $(Y, R(Y))$, but induce different values of $Q^\pi(h_t, y'_t)$, different baselines $V^\pi(h_t)$, and hence different action advantages in Equation (1). No estimator that observes only $(Y, R(Y))$ can distinguish the two environments, so it cannot recover the token-specific credits in both. $\square$

This proposition does not imply that outcome policy gradients are invalid: terminal Monte Carlo returns can be unbiased after averaging over sufficiently many trajectories. The limitation is finer-grained. Within one long response, the same outcome scalar cannot distinguish a useful reasoning step from a detour, a recoverable mistake, or a token whose effect is neutral. Outcome feedback is therefore reliable but sparse and high variance as a token-level learning signal.

A Globally Strong Teacher Can Be Locally Unreliable

Let $J(\pi; x)$ denote the success probability of policy π on prompt x . OPD commonly assumes a teacher that is stronger on average:

$$\mathbb{E}_x[J(\pi_T; x)] > \mathbb{E}_x[J(\pi_S; x)]. \quad (3)$$

This aggregate condition does not imply superiority at every prompt or every prefix visited by the student.

Proposition 2. A teacher can satisfy Equation (3) while providing an outcome-conflicting local preference on a student-generated prefix.

Proof. Consider two prompt types. Prompt x_1 occurs with probability 0.9, and prompt x_2 with probability 0.1. Let the student’s success probability be 0.5 on both prompts. Let the teacher’s success probabilities be 0.9 on x_1 and 0.1 on x_2 . The teacher is globally stronger because

$$0.9 \times 0.9 + 0.1 \times 0.1 = 0.82 > 0.5. \quad (4)$$

However, on x_2 the teacher is locally worse than the student. In a two-action realization, the teacher can assign higher probability to the incorrect action at a prefix of x_2 . The corresponding teacher–student log-probability gap then favors an action that decreases conditional success. Thus global teacher superiority does not guarantee local outcome alignment on every student-visited prefix. $\square$

The issue becomes more pronounced for long reasoning traces. OPD queries the teacher on prefixes generated by the student, including prefixes that may be rare under the teacher. Although the teacher always defines a conditional distribution at such states, its local logit difference need not be a reliable direction for improving the final verifier outcome.

Implication for POV-OPD

The two propositions expose a complementary gap. Outcome-only learning supplies a trustworthy final label but cannot resolve token-level credit from a single rollout. Unvalidated OPD supplies dense local information but cannot guarantee that this information is outcome-aligned at every student-generated prefix. POV-OPD therefore does not replace one signal with the other. It uses final correctness to validate the direction of dense teacher preferences, and uses sustained teacher excess surprisal as a conservative indicator of prefix drift. The next section describes how these two validation signals calibrate the original dense OPD advantage.

Method

Problem Setup

Let x be a prompt, π_θ the student policy, and π_T a fixed teacher policy. The student samples a response $y_i = (y_{i,1}, \dots, y_{i,T_i})$. At each response position t , the teacher and student score the sampled token $y_{i,t}$ on the student-visited prefix $(x, y_{i,<t})$. We collect the quantities used throughout the method as

$$\begin{aligned} y_i &\sim \pi_\theta(\cdot \mid x), \\ \ell_{i,t}^T &= \log \pi_T(y_{i,t} \mid x, y_{i,<t}), \\ \ell_{i,t}^\theta &= \log \pi_\theta(y_{i,t} \mid x, y_{i,<t}), \\ D_{i,t} &= \ell_{i,t}^T - \ell_{i,t}^\theta, \\ z_i &= 21[R_i > 1/2] - 1. \end{aligned} \quad (5)$$

where R_i is the verifier outcome. The teacher–student log-probability gap $D_{i,t}$ is the standard dense OPD advantage for the sampled token. The signed label z_i equals 1 for a correct response and -1 for an incorrect response.

From GRPO and OPD to POV-OPD

Figure 1 summarizes the three reward constructions. Given a group of responses to the same prompt, GRPO normalizes the terminal reward and broadcasts one trajectory-level advantage to every valid response token:

$$A_{i,t}^{\text{GRPO}} = \frac{R_i - \mu(x_i)}{\sigma(x_i) + \epsilon}, \quad 1 \leq t \leq T_i, \quad (6)$$

where $\mu(x_i)$ and $\sigma(x_i)$ are computed within the response group. GRPO therefore provides reliable but coarse outcome feedback. OPD instead provides the dense signal $D_{i,t}$, but does not verify whether each local teacher preference agrees with the final result or remains reliable on the student-generated prefix.

POV-OPD retains the dense OPD signal and calibrates it using outcome consistency, prefix reliability, and teacher support:

$$A_{i,t}^{\text{POV}} = D_{i,t} w_{i,t}^{\text{out}} w_{i,t}^{\text{pre}} w_{i,t}^{\text{sup}}. \quad (7)$$

Here w^{out} validates the direction of local supervision against the final outcome, w^{pre} attenuates supervision after sustained prefix drift, and w^{sup} mildly boosts aligned sampled tokens that remain well supported by the teacher.

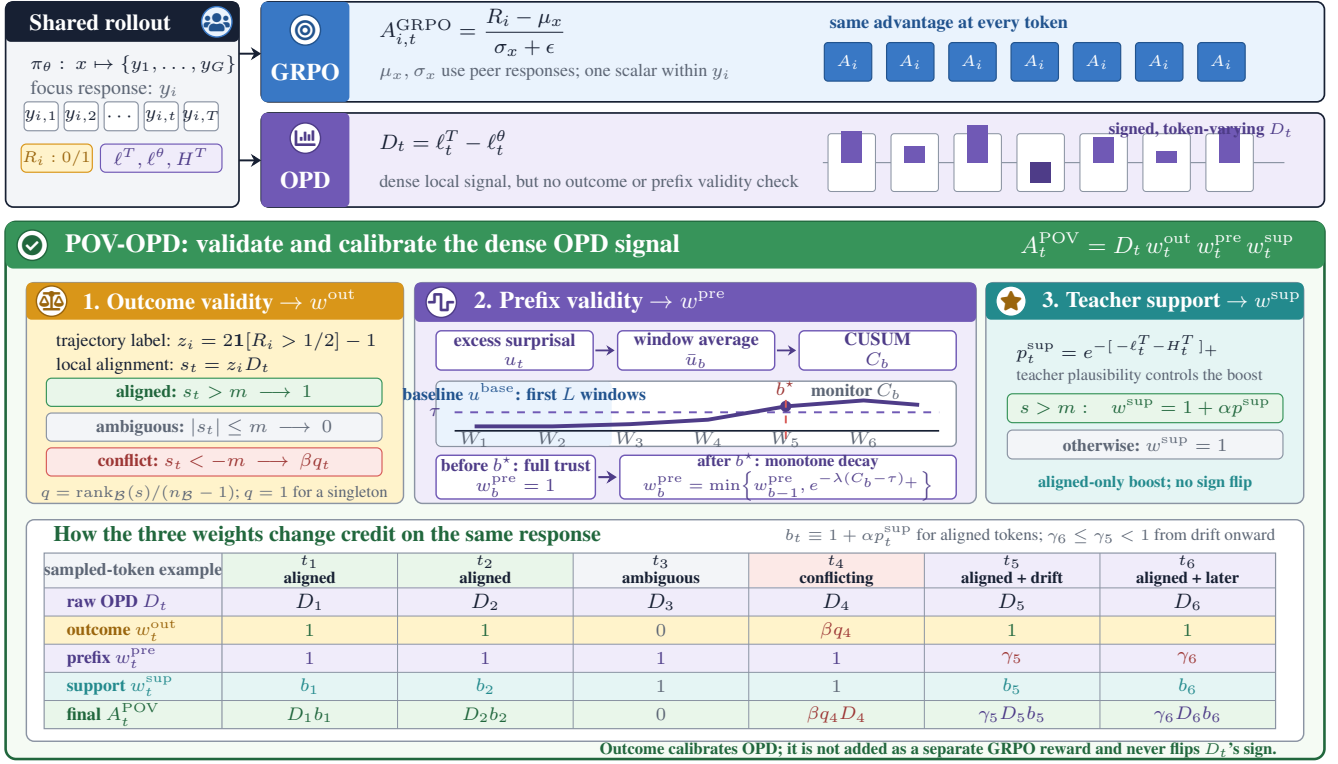


Figure 1: Credit assignment on the same student rollout. GRPO group-normalizes the terminal verifier reward and broadcasts one scalar within each trajectory, whereas OPD provides signed dense teacher–student gaps for sampled tokens without validating them. POV-OPD turns the terminal result into a local outcome-validity weight, detects sustained prefix drift from windowed teacher excess surprisal using CUSUM, and boosts only aligned sampled tokens that remain well supported by the teacher. The example at the bottom shows how the three nonnegative weights preserve, remove, cap, or decay each dense OPD signal without reversing its sign.

The worked example at the bottom of Figure 1 makes the calibration explicit: aligned dense credit is preserved and may be boosted, ambiguous credit is removed, anti-aligned credit is capped by βq , and credit after detected prefix drift is monotonically attenuated. The terminal outcome is therefore not added as a separate GRPO advantage and never reverses the sign of $D_{i,t}$; it only determines nonnegative validation weights.

Outcome Validation

Outcome validation signs each teacher–student gap by the trajectory result. We directly define the signed alignment score and its aligned and anti-aligned sets as

$$\begin{aligned} s_{i,t} &= z_i D_{i,t}, \\ \mathcal{A}_i &= \{t : s_{i,t} > m\}, \\ \mathcal{B}_i &= \{t : s_{i,t} < -m\}, \end{aligned} \quad (8)$$

where $m \geq 0$ is a tolerance margin and sampled tokens with $|s_{i,t}| \leq m$ are ambiguous. On a correct trajectory, a positive score means that the teacher assigns the sampled token higher probability than the student. On an incorrect trajectory, the sign is reversed, so teacher preferences that would

reinforce the failed behavior are treated as conflicting evidence.

A hard filter retains only $\mathcal{A}_i$, but may be brittle when either the verifier or a local teacher preference is noisy. We therefore retain aligned tokens fully and assign bounded rank weights only within the anti-aligned set. Let $n_i = |\mathcal{B}_i|$ and rank anti-aligned scores in ascending order, from 0 to $n_i - 1$. Then

$$\begin{aligned} q_{i,t} &= \begin{cases} 1, & n_i = 1, \\ \frac{\text{rank}_{\mathcal{B}_i}(s_{i,t})}{n_i - 1}, & n_i > 1, \end{cases} \\ w_{i,t}^{\text{out}} &= \begin{cases} 1, & t \in \mathcal{A}_i, \\ \beta q_{i,t}, & t \in \mathcal{B}_i, \\ 0, & |s_{i,t}| \leq m, \end{cases} \end{aligned} \quad (9)$$

where $\beta \in [0, 1]$ caps the retained strength of outcome-conflicting supervision. Anti-aligned tokens closer to the margin receive larger weights, while aligned tokens are never down-weighted. Setting $\beta = 0$ recovers hard outcome filtering.

Prefix Validation

Outcome validation checks the direction of a local preference; prefix validation checks whether the teacher remains well supported along the student trajectory. We compare the surprisal of the sampled token with the teacher entropy at the same prefix:

$$H_{i,t}^T = - \sum_a \pi_T(a \mid x, y_{i,<t}) \log \pi_T(a \mid x, y_{i,<t}), \quad (10)$$

$$u_{i,t} = - \log \pi_T(y_{i,t} \mid x, y_{i,<t}) - H_{i,t}^T.$$

A positive $u_{i,t}$ means that the sampled token is more surprising to the teacher than a typical token under the teacher distribution. Isolated high-surprisal tokens need not indicate failure, so we detect sustained rather than instantaneous drift.

We divide the valid response into M_i consecutive windows of W tokens and denote the mean excess surprisal in window b by $\bar{u}_{i,b}$. The first $L_i = \min(L, M_i)$ windows form a response-specific baseline. Baseline estimation, one-sided CUSUM accumulation, and horizon detection are

$$u_i^{\text{base}} = \frac{1}{L_i} \sum_{b=1}^{L_i} \bar{u}_{i,b}, \quad C_{i,L_i} = 0,$$

$$C_{i,b} = \max(0, C_{i,b-1} + \bar{u}_{i,b} - u_i^{\text{base}}), \quad b > L_i,$$

$$b_i^* = \min\{b > L_i : C_{i,b} > \tau\}, \quad (11)$$

where τ is the detection threshold. If the threshold is never crossed, all prefix weights remain one.

After detection, the window weight decays monotonically:

$$\eta_{i,b} = \exp[-\lambda \max(C_{i,b} - \tau, 0)],$$

$$w_{i,b}^{\text{pre}} = \begin{cases} 1, & b < b_i^*, \\ \min(w_{i,b-1}^{\text{pre}}, \eta_{i,b}), & b \geq b_i^*. \end{cases} \quad (12)$$

Each window weight is assigned to its valid tokens. The monotone recursion prevents later fluctuations from restoring full teacher trust after sustained prefix drift has already been detected.

Teacher-Supported Boost

Among outcome-aligned sampled tokens, POV-OPD gives a small boost to those that are also plausible under the teacher distribution. Eliminating an unnecessary intermediate surprisal variable, we define support and its weight directly as

$$p_{i,t}^{\text{sup}} = \exp[-\max(-\ell_{i,t}^T - H_{i,t}^T, 0)], \quad (13)$$

$$w_{i,t}^{\text{sup}} = 1 + \alpha p_{i,t}^{\text{sup}} \mathbf{1}[t \in \mathcal{A}_i],$$

where $\alpha \geq 0$ controls the maximum boost. Support equals one when the sampled token is no more surprising than the teacher entropy and decays otherwise. Anti-aligned and ambiguous tokens are not boosted.

Optimization

We substitute Equation (7) for the original OPD advantage in the same clipped PPO-style update. Writing the policy

ratio once and using it directly in the loss gives

$$r_{i,t}(\theta) = \frac{\pi_\theta(y_{i,t} \mid x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t} \mid x, y_{i,<t})},$$

$$\mathcal{L}_{\text{POV}} = - \sum_{i,t} \min \left\{ r_{i,t}(\theta) A_{i,t}^{\text{POV}}, \text{clip}_{[1-\epsilon, 1+\epsilon]}(r_{i,t}(\theta)) A_{i,t}^{\text{POV}} \right\}. \quad (14)$$

All validation weights are computed without gradient and held fixed during the actor update. POV-OPD therefore changes the credit assigned to each sampled token while preserving the rollout generation, teacher scoring, and actor optimization pipeline of OPD.

Experiments

We evaluate whether prefix–outcome validation corrects unreliable token-level OPD gradients and thereby improves on-policy distillation for reasoning. We compare **POV-OPD** with two complementary baselines: **GRPO**, which uses only sparse verifier feedback, and **OPD**, which uses dense teacher–student logit differences without outcome or prefix validation. We conduct experiments under four student–teacher configurations, spanning different model families and capacity gaps. Within each configuration, all methods use the same student initialization, training data, rollout budget, and optimization hyperparameters whenever applicable. Thus, the comparison isolates the contribution of validating dense teacher supervision.

Experimental Setup

Training. We train on DAPO-Math-17K (Yu et al. 2025) and consider four configurations: (1) Qwen3-1.7B distilled from Qwen3-4B Thinking (Yang et al. 2025); (2) DeepSeek-R1-Distill-Qwen-1.5B (DeepSeek-AI 2025) distilled from JustRL-1.5B (He et al. 2025a); (3) DeepSeek-R1-Distill-Qwen-1.5B distilled from Skywork-OR1-Math-7B (He et al. 2025b); and (4) DeepSeek-R1-Distill-Qwen-7B distilled from Skywork-OR1-Math-7B. For each prompt, the student samples on-policy responses and is evaluated by a rule-based mathematical verifier. GRPO uses the resulting outcome reward to construct group-relative advantages. OPD uses the teacher–student log-probability gap as a dense token-level signal. POV-OPD uses the same dense OPD signal, but outcome validation distinguishes outcome-aligned from anti-aligned teacher preferences and prefix validation attenuates supervision after sustained teacher-surprisal drift. In each configuration, GRPO uses the same student initialization but does not query the teacher; the listed teacher is used only by OPD and POV-OPD.

Evaluation. We evaluate on four reasoning benchmarks: AMC23 (AMC), AIME 2024 (AIME24), and AIME 2025 (AIME25) from the MAA competition series (Mathematical Association of America 2025), and GPQA (Rein et al. 2023). For each problem, we independently sample eight responses from the evaluated policy. $\text{Avg}@8$ is the mean correctness over the eight samples, whereas $\text{Best}@8$ reports whether at least one of the eight samples is correct, averaged over problems. Higher values are better for both metrics. We

Table 1: Main results and compact mechanism audit. Panel (a) compares four student–teacher configurations; entries are percentages. Panels (b)–(d) use DeepSeek-R1-Distill-Qwen-1.5B with JustRL-1.5B. O/P/S denote outcome validation, prefix validation, and teacher-supported boosting. Panel (c) compares gradients on rollout set A with an independent outcome-gradient reference from rollout set B on the same prompts. “Neg.” and “Zero” denote negative-cosine and near-zero method-gradient rates, respectively.

Student / Teacher	Method	AMC		AIME24		AIME25		GPQA	
		Avg@8	Best@8	Avg@8	Best@8	Avg@8	Best@8	Avg@8	Best@8
Qwen3-1.7B / Qwen3-4B-T	GRPO	—	—	—	—	—	—	—	—
	OPD	—	—	—	—	—	—	—	—
	POV-OPD	—	—	—	—	—	—	—	—
DS-R1-Qwen-1.5B / JustRL-1.5B	GRPO	—	—	—	—	—	—	—	—
	OPD	—	—	—	—	—	—	—	—
	POV-OPD	—	—	—	—	—	—	—	—
DS-R1-Qwen-1.5B / Skywork-OR1-7B	GRPO	—	—	—	—	—	—	—	—
	OPD	—	—	—	—	—	—	—	—
	POV-OPD	—	—	—	—	—	—	—	—
DS-R1-Qwen-7B / Skywork-OR1-7B	GRPO	—	—	—	—	—	—	—	—
	OPD	—	—	—	—	—	—	—	—
	POV-OPD	—	—	—	—	—	—	—	—

(b) Component ablation					
Variant	O	P	S	Macro Avg@8	Macro Best@8
OPD	—	—	—	—	—
+ Outcome	✓	—	—	—	—
+ Prefix	—	✓	—	—	—
+ Outcome + Prefix	✓	✓	—	—	—
POV-OPD	✓	✓	✓	—	—

(c) Independent-reference gradient audit				
Signal on A	Cosine $\uparrow$	Neg. (%) $\downarrow$	Zero (%) $\downarrow$	Rel. norm
Outcome-PG	—	—	—	—
Raw OPD	—	—	—	—
POV-OPD	—	—	—	—

(d) Does b^* identify less outcome-compatible OPD gradients?					
Group	Conflict pre (%)	Conflict post (%)	Δ (pp)	Mean cos. pre	Mean cos. post
Triggered	—	—	—	—	—
Position-matched control	—	—	—	—	—

use the same decoding budget, prompt template, and rule-based answer verifier for all compared methods within each benchmark.

Main Results

Table 1(a) reports the main comparison across all four student–teacher configurations. The table jointly measures average reasoning reliability and the benefit of repeated sampling. Avg@8 measures the quality of a typical sampled response, while Best@8 measures the attainable performance when a small response budget is available. We report both because a method can improve the best sampled trajectory while degrading the average trajectory quality, which would be undesirable for practical reasoning systems.

Reading the comparison. The most direct evidence for POV-OPD is a consistent improvement over OPD on Avg@8: this would show that validating dense supervision improves the quality of typical on-policy trajectories rather than merely increasing sampling variance. Gains on Best@8 additionally indicate that the method preserves or improves exploration of successful solution paths. In contrast, a gain over GRPO demonstrates the value of dense teacher guidance beyond sparse outcome feedback. We further analyze how outcome and prefix validation reshape unreliable OPD gradients and isolate the contribution of each component in the following ablations.

Component Ablation and Gradient Audit

Panels (b)–(d) of Table 1 use DeepSeek-R1-Distill-Qwen-1.5B as the student and JustRL-1.5B as the teacher. The component ablation changes only the validation weights. Outcome-only disables prefix weighting and sets $\alpha = 0$; prefix-only retains every raw OPD token, applies only w^{pre} , and also sets $\alpha = 0$; outcome+prefix keeps both validation weights but omits the support boost. The final two rows therefore isolate the effect of w^{sup} . Macro Avg@8 and Best@8 give equal weight to each of the four benchmarks in the main evaluation.

Gradient protocol. At each pre-specified frozen student checkpoint, we use the same held-out prompts to sample two independent response groups, A_j and B_j . Method gradients are computed on A_j , while B_j supplies an independent outcome-policy-gradient reference $g_{j,B}^{\text{out}}$. Outcome-PG on A_j measures the agreement between two independent outcome estimators. All advantages and validation weights are detached, and every objective uses the same trainable parameters, token masking, and loss normalization. We first define one common audit set $\mathcal{J} = \{j : \|g_{j,B}^{\text{out}}\|_2 \geq \epsilon_{\text{ref}}\}$ using only the reference norm, and use this set for every method. For $X \in \{\text{out}, \text{OPD}, \text{POV}\}$, let $\mathcal{J}_X \subseteq \mathcal{J}$ contain batches whose method gradient also exceeds a pre-specified

numerical threshold. We report

$$c_j^X = \frac{\langle g_{j,A}^X, g_{j,B}^{\text{out}} \rangle}{\|g_{j,A}^X\|_2 \|g_{j,B}^{\text{out}}\|_2},$$

$$p_-^X = \frac{1}{|\mathcal{J}_X|} \sum_{j \in \mathcal{J}_X} \mathbf{1}[c_j^X < 0], \quad z_0^X = 1 - \frac{|\mathcal{J}_X|}{|\mathcal{J}|}. \quad (15)$$

Cosine is averaged over $\mathcal{J}_X$ per audit batch rather than computed after averaging gradients. Near-zero method gradients are not silently discarded: their rate z_0^X is reported, while the common reference-valid count is $|\mathcal{J}| = -$. We also report the batch-wise mean of $\|g_{j,A}^X\| / (\|g_{j,B}^{\text{out}}\| + \epsilon)$ over all $j \in \mathcal{J}$, including method-gradient zeros, so that an apparent alignment gain cannot be attributed merely to shrinking the update toward zero.

Component effects. Outcome validation changes Macro Avg@8 from $-$ to $-$, while prefix validation alone obtains $-$. Combining them yields $-$, and adding teacher-supported boosting gives $-$. These comparisons respectively test outcome correction, prefix correction, their complementarity, and the incremental value of support; a component is treated as beneficial only when its gain exceeds the evaluation uncertainty.

Overall gradient correction. Against the independent outcome reference, raw OPD obtains cosine $--$ with $--\%$ negative-cosine batches, whereas POV-OPD obtains $--$ with $--\%$. The corresponding relative norms are $--$ and $--$. This audit measures agreement with an outcome-based optimization direction, not access to a ground-truth token gradient; it is therefore interpreted as a mechanism check rather than a proof of token correctness.

Before and after the detected horizon. For a window $W_{i,b}$, define its raw OPD update as $g_{i,b}^{\text{OPD}} = |W_{i,b}|^{-1} \sum_{t \in W_{i,b}} D_{i,t} \nabla_{\theta} \log \pi_{\theta}(y_{i,t} | h_{i,t})$. It is conflicting iff $\langle g_{i,b}^{\text{OPD}}, g_{j,B}^{\text{out}} \rangle < 0$. Panel (d) reports the equal-window mean of the corresponding cosines, separately before b^* and over the trigger window and all later windows. Each triggered trajectory is matched to an untriggered rollout of the same prompt and outcome class with similar length; the control uses a pseudo-boundary at the same normalized position. CUSUM triggers on $--\%$ of audited trajectories, leaving $--$ matched pairs. In triggered trajectories, the conflict rate changes from $--\%$ before b^* to $--\%$ after it, compared with $--\%$ to $--\%$ for the position-matched control. The difference-in-differences is $--$ percentage points. Evidence for localization requires this value to be positive with a 95% confidence interval excluding zero and a larger post-horizon cosine deterioration than in the matched control. Because prefix weights are positive constants within each window, they cannot alter that window’s direction, conflict indicator, or cosine. Panel (d) therefore tests whether b^* localizes raw OPD regions that are less outcome-compatible on average; the method attenuates the mass assigned to those post-horizon regions, while aggregate gradient correction is tested separately in panel (c).

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